

Report 04

Core Agent Development: Weather, Ocean & Geo Agents

Project:	JalJeev — Agentic Marine Intelligence System
Phase:	Phase 0, Week 2 (Sep 1–7, 2026)
Focus:	Building the three foundational deterministic agents and multi-source data fusion

1. Objective

Build the three core data-gathering agents (Weather, Ocean, Geo) that form the input layer of the entire decision pipeline. Every agent is **fully deterministic** — pure Python, no LLM. The LLM only touches the system at the orchestration layer for intent parsing and answer phrasing.

2. Architectural Principle: Agents vs. Adapters

Layer	Responsibility	Returns
Adapter	Talk to one external source; timeout, retry, error handling	Raw data with {status, source, data}
Agent	Orchestrate multiple adapters, compute derived values, deduplicate	Types, Python schema

Adapters never import each other. Agents never talk to external sources directly. This separation means a new data source is a new adapter file — no agent code changes.

3. Weather Agent

3.1 Priority Chain (per PRD Section 7.3)

IMD (primary, pending) → Open-Meteo Forecast (fallback, live) → ERA5 (historical, later)

3.2 Data Fields Collected

Field	Source	Safety Relevance
Sustained wind speed (m/s)	IMD or Open-Meteo	Small craft limit >12 m/s
Wind gusts (m/s)	Open-Meteo	Sudden squall indicator
Wind direction (deg)	IMD or Open-Meteo	Sail risk, route planning
Precipitation probability (%)	Open-Meteo	Visibility, deck safety
Atmospheric pressure (hPa)	Open-Meteo	Storm precursor
Pressure trend (hPa/3hr)	Open-Meteo	Rapid drop = squall warning
Temperature / Dewpoint (°C)	Open-Meteo	Fog risk
Fog risk (bool)	Derived	Visibility hazard
Cyclone active (bool)	IMD (pending)	Disaster mode trigger

3.3 Auto-Activation Design

When `IMD_API_KEY` is dropped into `.env`, the Weather Agent **automatically** starts using IMD as primary with zero code changes elsewhere. The IMD adapter itself reports "unavailable" until then, and the agent moves to the next source.

3.4 Honest Gap Reporting

When IMD is unavailable, the agent explicitly records gaps in its `WeatherState.missing` list. These gaps propagate to the Evidence Agent, the UI, and the answer text.

4. Ocean Agent

4.1 The Most Complex Agent — 6 Concurrent Source Calls

The Ocean Agent runs **six** adapter calls simultaneously via `asyncio.gather`:

```
incois_result, om_marine_result, noaa_result, copernicus_result, bgc_result, tide_result = await asyncio.gather(
    incois_adapter.get_incois_wave_forecast(lat, lon),
    open_meteo_adapter.get_marine_forecast(lat, lon),
    noaa_oisst_adapter.get_sst(lat, lon),
    copernicus_adapter.get_physics_subset(lat, lon),
    copernicus_adapter.get_bgc_subset(lat, lon),
    tide_adapter.get_tide(lat, lon),
)
```

Why concurrent: Running these 6 calls sequentially would let worst-case latencies add up past a minute. Concurrently, total latency is bounded by the slowest single source (~10-25s), not the sum.

4.2 Multi-Source SST Consensus

SST from NOAA OISST and Copernicus Marine are averaged. If they diverge by more than 1.0°C, a disagreement flag fires — this is the uncertainty engine in action.

4.3 Wave/Swell Separation and Cross-Sea Detection

Open-Meteo Marine provides separate swell and wind-wave components. The agent detects **cross-swell** — when two wave trains arrive from meaningfully different directions (>40° angular difference, both >0.5m) — because this is more dangerous than either alone.

4.4 Tide Prediction (EOT20)

The EOT20 empirical ocean tide model, via `pyTMD`, predicts astronomical tide only (not storm surge). Returns tide height in metres and state (rising/falling/high/low).

4.5 Chlorophyll & HAB Proxy

The BGC subsystem from Copernicus Marine provides chlorophyll-a concentration. A coarse harmful algal bloom threshold (5.0 mg/m³) is applied as a "worth a second look" flag, explicitly documented as not a calibrated regional threshold.

5. Geo Agent

5.1 PostGIS-Powered Spatial Queries

Query	Data Source	SQL Function
Is this point inside the Indian EEZ?	Marine Regions v12	ST_Contains(eez_geom, point)
Is this point inside an MPA?	WDPA / Protected Planet	ST_Contains(mpa_geom, point)
Which is the nearest port?	Ports table	ST_Distance + ORDER BY
What is the water depth here?	GEBCO bathymetry tiles	Numpy array lookup
Is this point on land?	GEBCO	Direct check (depth = 0 or positive)

5.2 Boundary Loading

EEZ and MPA boundaries were loaded into PostGIS using scripts/load_boundaries.py:

- Marine Regions World EEZ v12 — GeoPackage → PostGIS
- WDPA marine protected areas — bbox-filtered to the Indian Ocean → PostGIS

6. Pydantic Output Schemas

Every agent returns a typed Pydantic model:

```
class WeatherState(BaseModel):
    wind_speed_ms: float | None = None
    wind_gust_ms: float | None = None
    sources_used: list[str] = []
    data_freshness: dict[str, str] = {}
    partial: bool = False
    missing: list[str] = []

class OceanState(BaseModel):
    significant_wave_height_m: float | None = None
    sst_c: float | None = None
    sst_sources: dict[str, float] = {}
    sst_disagreement: float | None = None

class GeoState(BaseModel):
    inside_indian_eez: bool = False
    inside_mpa: bool = False
    depth_m: float | None = None
    is_land: bool = False
```

Every field defaults to None, never to a plausible-looking zero. A missing value and a true zero are fundamentally different things — the system must never confuse them.

7. Additional Adapters Built This Week

Adapter	Source	Auth	Data
open_meteo_adapter.py	Open-Meteo Forecast + Marine	None	Wind, wave, swell, pressure, precipitation
noaa_oisst_adapter.py	NOAA ERDDAP	None	Sea surface temperature
copernicus_adapter.py	Copernicus Marine	Username/password	Physics (SST, currents), BGC (chlorophyll)
tide_adapter.py	Local EOT20 model via pyTMD	None	Astronomical tide prediction
mosdac_adapter.py	ISRO MOSDAC Download API	Username/password	EOS-06 OCM-3 chlorophyll

8. Week 2 Deliverables

Deliverable	Status
Weather Agent (deterministic, IMD fallback chain)	■
Ocean Agent (6 concurrent sources, SST consensus)	■
Geo Agent (PostGIS EEZ/MPA/depth queries)	■
5 new data adapters	■
Pydantic schemas for all agent outputs	■
EEZ + MPA boundaries loaded into PostGIS	■

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